

Dado las siguientes ecuaciones en diferencias de los siguientes sistemas, que representan un filtro de media móvil:

a) $y(n) = x(n-3) + x(n-2) + x(n-1) + x(n)$

b) $y(n) = x(n-4) + x(n-3) + x(n-2) + x(n-1) + x(n)$

c) $y(n) = x(n) - x(n-1)$

d) $y(n) = x(n) - x(n-2)$

Se pide:

1) Hallar $T(z) = \frac{Y(z)}{X(z)}$

$$X(z) = \sum_{n=-\infty}^{+\infty} x(n) z^{-n}$$

$$Y(z) = \sum y(n) z^{-n} = \sum_{\substack{m=n-n_0 \\ n=m+n_0}} x(n-n_0) z^{-n} = \sum x(m) z^{-(m+n_0)}$$

$$Y(z) = \underbrace{\sum_m x(m) \cdot z^{-m}}_{X(z)} \cdot \underbrace{z^{-n_0}}_{\text{offset } z^{-n_0} \text{ den-shift, solve in } Y(z) \text{ by } z^{-n_0}}$$

$$Y(z) = X(z) z^{-n_0}$$

$$T(z) = \frac{Y(z)}{X(z)}$$

a) $y(n) = x(n-3) + x(n-2) + x(n-1) + x(n)$

$$Y(z) = X(z) z^{-3} + X(z) z^{-2} + X(z) z^{-1} + X(z) z^0$$

$$Y(z) = X(z) (z^{-3} + z^{-2} + z^{-1} + 1)$$

$$\frac{Y(z)}{X(z)} = T(z) = \frac{1}{z^3} (1 + z + z^2 + z^3)$$

$$z_1 = -1, \quad z_2 = -j, \quad z_3 = j$$

$$T(z) = \frac{(z+1)(z-j)(z+j)}{z^3}$$



Para mostrar Frec

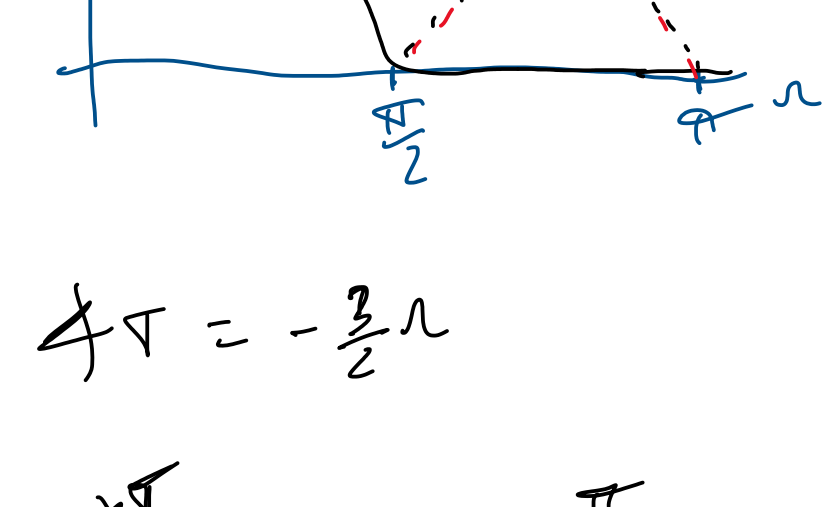
$$T(j\omega) \Big|_{z=e^{j\omega}}$$

$$T(j\omega) = \frac{e^0 + e^{j\omega} + e^{j2\omega} + e^{j3\omega}}{e^{j3\omega}} = e^{-j3\omega} (e^{j3\omega} + e^{j2\omega} + e^{j\omega} + 1)$$

$$= e^{-j3\omega} (e^{j3\omega} + e^{j2\omega} + e^{j\omega} + 1)$$

$$T(j\omega) = \underbrace{e^{-j3\omega}}_{\cancel{}} \underbrace{(2 \cos(\frac{\omega}{2}) + 2 \cos(\frac{3\omega}{2}))}_{|T|}$$

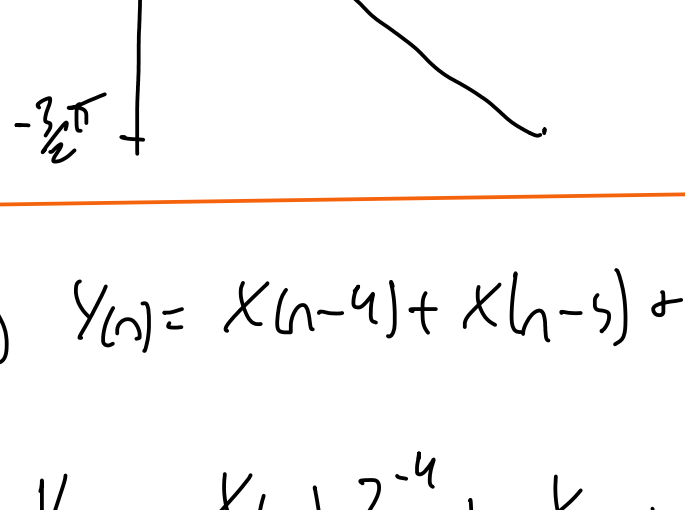
$$T(0) = 4, \quad T(\frac{\pi}{2}) = 2 \cdot \frac{\sqrt{2}}{2} + 2 \cdot \frac{\sqrt{2}}{2} = 0, \quad T(\pi) = 2 \cdot 0 + 2 \cdot 0 = 0$$



Filtro pasa bajo

$$\angle T = -\frac{3}{2}\omega$$

$$\angle T(0) = 0$$



$$\angle T(\pi) = -\frac{3}{2}\pi$$

b) $y(n) = x(n-4) + x(n-3) + x(n-2) + x(n-1) + x(n)$

$$Y(z) = X(z) z^{-4} + X(z) z^{-3} + X(z) z^{-2} + X(z) z^{-1} + X(z) z^0$$

$$\frac{Y(z)}{X(z)} = z^4 + z^3 + z^2 + z^1 + 1 = T(z)$$

$$\frac{1}{z^4} (1 + z + z^2 + z^3 + z^4)$$

1/2 new poles added
2 new zeros

$$1 + z^1 + z^2 + \dots + z^{n-1} = \frac{1-z^n}{1-z}$$

$$\frac{1-z^5}{1-z} \rightarrow \text{cancel low cancel}$$

$$1 - z^5 = 0$$

$$1 = z^5$$

with $K=1, 2, 3, 4$

$K=0$ the same

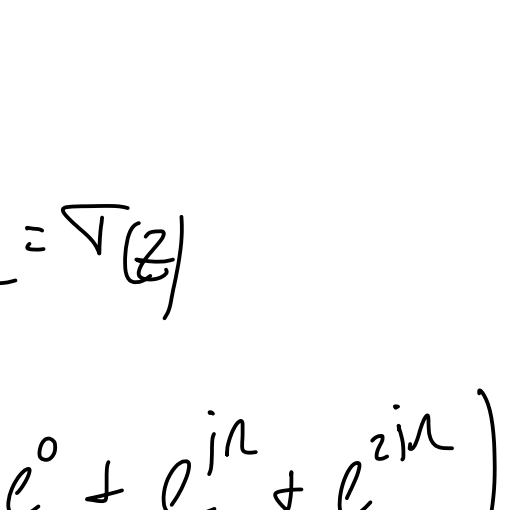
$$e^{j\frac{2\pi K}{5}}$$

$$z_{(K=1)} = e^{j\frac{2\pi}{5}} \rightarrow 0.4$$

$$z_{(K=2)} = e^{j\frac{4\pi}{5}} \rightarrow 0.6$$

$$z_{(K=3)} = e^{j\frac{6\pi}{5}} \rightarrow 1.2$$

$$z_{(K=4)} = e^{j\frac{8\pi}{5}} \rightarrow 1.6$$



low pass Freq

$$(z^4 + z^3 + z^2 + z^1 + 1) \frac{1}{z^4} = T(z)$$

$$T(j\omega) = \frac{e^{j4\omega} (e^{-j4\omega} + e^{-j3\omega} + e^{-j2\omega} + e^{-j\omega} + 1)}{e^{j4\omega}}$$

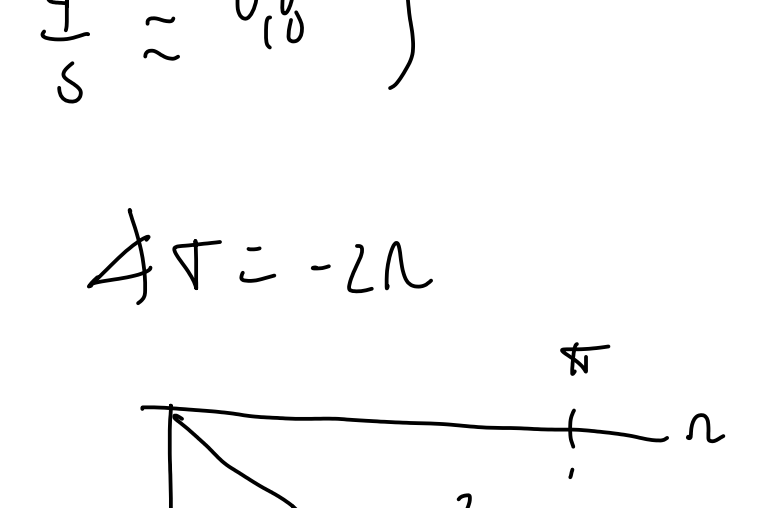
$$= e^{-j4\omega} (e^{j4\omega} + e^{j3\omega} + e^{j2\omega} + e^{j\omega} + 1)$$

$$T(j\omega) = \underbrace{e^{-j4\omega}}_{\cancel{}} \underbrace{(2 \cos(2\omega) + 1 + 2 \cos(\omega))}_{|T|}$$

$$T(0) = 2 \cdot \cos(0) + 1 + 2 \cos(0) = 5 = |T(0)|$$

$$T(\frac{\pi}{2}) = 2 \cdot \cos(\pi) + 1 + 2 \cdot \cos(\frac{\pi}{2}) = 1 = |T(\frac{\pi}{2})|$$

$$T(\pi) = 2 \cdot \cos(2\pi) + 1 + 2 \cos(\pi) = 1 = |T(\pi)|$$



$$T(\frac{3}{2}\pi) = T(\frac{4}{5}\pi) = 0$$

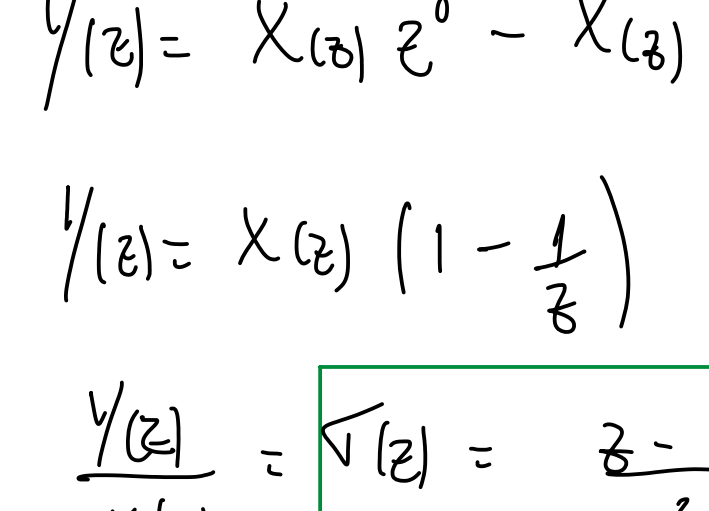
$$\frac{2}{5} \approx 0.4$$

$$\frac{4}{5} \approx 0.8$$

Filtro pasa bajo

$$\angle T = -2\omega$$

$$\angle T(0) = 0$$



$$\angle T(\pi) = -2\pi$$

c) $y(n) = x(n) - x(n-1)$

$$Y(z) = X(z) z^0 - X(z) z^{-1}$$

$$Y(z) = X(z) (1 - \frac{1}{z})$$

$$\frac{Y(z)}{X(z)} = T(z) = \frac{z-1}{z}$$

poles at $z=0 \rightarrow$ (simple)

zeros at $z=1 \rightarrow$

$$\boxed{z=1}$$

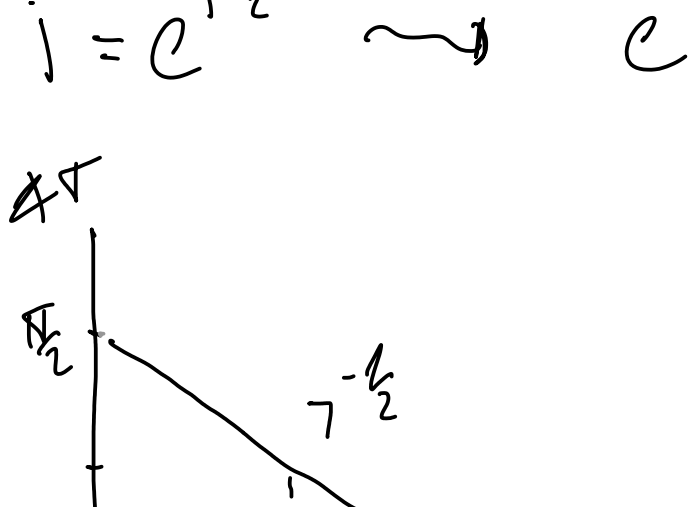
$$T(j\omega) = \frac{e^{j\omega} - 1}{e^{j\omega}} = \frac{e^{j\omega} - e^{j0}}{e^{j\omega}} = \frac{e^{j\frac{\omega}{2}} (e^{j\frac{\omega}{2}} - e^{-j\frac{\omega}{2}})}{e^{j\omega}}$$

Identity: $\sin$

$$2j \sin(\theta) = e^{j\theta} - e^{-j\theta}$$

$$= e^{-j\frac{\omega}{2}} (2j \sin(\frac{\omega}{2})) = T(j\omega)$$

$$2 \sin(\frac{\omega}{2}) = |T|$$



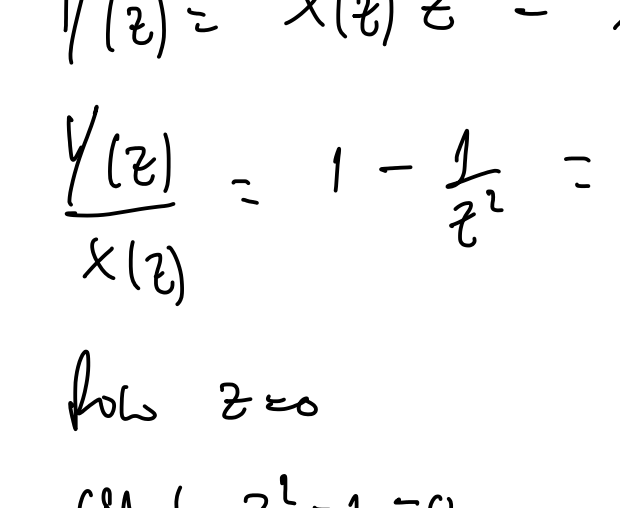
$$T(0) = 0$$

$$T(\frac{\pi}{2}) = \frac{\sqrt{2}}{2} \cdot 2 = \sqrt{2}$$

$$T(\pi) = 2$$

Filtro pasa alto

$$j = e^{j\frac{\pi}{2}} \rightarrow e^{-j\frac{\pi}{2}} \cdot e^{j\frac{\pi}{2}} = e^{j(\frac{\pi}{2} - \frac{\pi}{2})} = \cancel{1}$$



$$\angle T = \frac{\pi}{2} - \frac{\omega}{2}$$

$$\angle T(0) = \frac{\pi}{2}$$

$$\angle T(\pi) = 0$$

d) $y(n) = x(n) - x(n-2)$

Filtro pass

$$Y(z) = X(z) z^0 - X(z) z^{-2}$$

$$\frac{Y(z)}{X(z)} = 1 - \frac{1}{z^2} = \frac{z^2 - 1}{z^2} = T(z)$$

poles $z=0$

$$\text{zeros } z^2 - 1 = 0$$

$$z^2 = 1$$

$$T(j\omega) = \frac{e^{j2\omega} - 1}{e^{j2\omega}} = \frac{e^{j\omega} (e^{j\omega} - e^{-j\omega})}{e^{j2\omega}} = e^{-j\omega} (2j \sin(\omega))$$

$$|T| = 2 \sin(\omega)$$

$$j e^{-j\omega} = e^{j\frac{\pi}{2}} e^{-j\omega} = e^{j(\frac{\pi}{2} - \omega)} = \cancel{1}$$



$$T(0) = 0$$

$$T(\frac{\pi}{2}) = 2$$

$$T(\pi) = 0$$

Filtro pasa banda

$$\angle T = \frac{\pi}{2} - \omega$$

$$\angle T(0) = \frac{\pi}{2}$$

$$\angle T(\pi) = -\frac{\pi}{2}$$

